

cli/src/main.rs · Code explanation (L250-317)

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What it does `prefetch_summary_queue` builds a limited queue of `PrefetchSummaryJob`s from a list of file changes. It filters out changes that should be ignored by `.gitignore`, removes "novadiff" documents, discards non-text files, then orders the remaining changes by depth, subtree weight, and path. The resulting jobs contain the original path, kind, depth, and the weight of the parent subtree.

How it works
1. Initial counts – `input_changes` records the total number of changes.
2. **Gitignore filtering*

- If `gitignore_roots` is provided, `filter_changes_for_prefetch_gitignore` returns a tuple (`after_gitignore`, `skipped_gitignore`); otherwise all changes are kept and `skipped_gitignore` is 0. 3. **Reserved docs filtering*
- `filter_out_novadiff_docs_changes` removes "novadiff" docs, yielding (`after_reserved`, `skipped_novadiff_docs`). 4. **Non-text filtering*
- When `gitignore_roots` is present, `filter_out_non_text_changes` removes non-text files, producing (`eligible`, `skipped_non_text`); otherwise `eligible` is `after_reserved`. 5. **Weight calculation*
- `eligible` paths are converted to Unix style (`replace('\\', '/')`) and `subtree_change_counts` builds a map of parent-directory weights. 6. **Sorting*
- `eligible` is sorted by:
 - path depth (`path_depth`),
 - weight of the parent directory (`weights[parent_dir]`),
 - lexicographic path order. 7. **Queue construction*
- Up to `limit` (at least 1) sorted changes are turned into `PrefetchSummaryJob`s, each carrying the original path, kind, depth, and the parent subtree weight. 8. **Return*
- A `PrefetchSummaryQueueOut` struct records the queue and all skip counts.

Notes for reviewers- The function assumes

`filter_changes_for_prefetch_gitignore`, `filter_out_novadiff_docs_changes`, `filter_out_non_text_changes`, `subtree_change_counts`, `path_depth`, and `parent_dir` are defined elsewhere; their behavior is inferred only from usage.-

The `limit.max(1)` guard ensures at least one job is considered, even if `limit` is zero.- All string manipulations use `replace('\\', '/')` to normalise Windows paths before sorting.- The returned struct contains both the raw counts and the derived queue, facilitating downstream diagnostics.